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The Ethics of Empathetic AI in Medicine

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***Abstract*—The expression of empathy is an important part of effective and humane medical care. Modern medicine faces a significant challenge in this area, at least in part due to the ever-increasing demands on clinicians' time. When empathic responses to patient emotions are lacking, the quality of care suffers, including a patient's ability to understand important information and willingness to trust their doctor. This article asks whether AI can help address this challenge. We approach the question by presenting and critically analyzing three novel scenarios that represent fundamentally different approaches, including (1) AI-powered "doctor" avatars directly interacting with patients, (2) AI editors and co-authors helping clinicians find the right words, and (3) AI simulated patients helping to provide communication training. In each case, we identify fundamental AI and medical ethics considerations, including those that warrant further consideration.**

***Index Terms*— empathy, medical empathy, AI, AI ethics, medical ethics**

I. INTRODUCTION

EMPATHY is a remarkable ability. It plays an important role in relationships—perhaps even a fundamental one in moral reasoning, moral development, and compassionate care—by sensitizing us to how others are feeling and thinking. Unfortunately, modern medicine has an empathy problem. Despite the proliferation of scholarly work and the availability of communication training programs over the past two decades, doctors often fail to express empathy, especially in stressful moments when patients and their families are struggling to hear bad news and make difficult decisions. Since empathy has been shown to enhance patients' understanding, adherence to a medical plan, and even how much they trust their medical team, falling short can negatively affect the quality of patient care [1].

It has been suggested that Artificial Intelligence (AI) might be able to help. That might sound like an ironic approach to the problem because doctors who struggle to express empathy can come across as robotic. They can present diagnoses and treatment options in an unemotional and detached manner that seems uncaring, as well as fail to sensitively acknowledge when patients are anxious, if not terrified. Yet researchers and healthcare professionals are increasingly considering the possibility that AI can help address medicine's empathy problem, which stems, at least in part, from difficult-to-change environmental and psychological factors. Among those factors are ever-increasing pressures to quickly see large numbers of patients, navigate often user-unfriendly electronic medical record systems, and address seemingly endless paperwork and a myriad of administrative duties. These taxing conditions ultimately lead to the epidemic of burnout [2].

In addition to these challenges, there is the increasingly recognized syndrome of "moral injury." Initially, it referred to the pain soldiers felt when they engaged in or observed behavior that contradicted their deeply held moral beliefs and values. But today, moral injury has become part of medical conversations. As one physician aptly notes [3], it also "describe[s] the feelings of guilt, sadness and defeat felt by health care professionals when we know what our patients need but can't provide it."

Crucially, none of the novel, AI-based proposals address the root causes of the problem—the deeply entrenched obstacles that prevent doctors from spending more quality time with their patients. And while each proposal is well-intentioned, the entire endeavor is fraught with risks. Not only is it unrealistic to expect AI to fix everything, but using it could also cause avoidable harm. Every decision to use cutting-edge technology when lower-risk options are available is a moral choice. At the same time, risk cuts both ways. Deciding that AI cannot help could leave us stuck with the current dysfunctional healthcare system, where patients often see doctors as robotic.

II. THREE TYPES OF EMPATHY AND THREE MEDICAL AI SCENARIOS

Since AI can potentially improve empathetic medical communication, we may be on the verge of creating new medical practices and engineering [4] new reactions to them that impact who and what people trust [5]. To weigh benefits, harms, and risks, we need to begin with a baseline understanding of how different aspects of empathy enable doctors to accomplish core tasks [6].

A doctor sees a child clinging tightly to a parent and immediately recognizes her anxiety about getting a scheduled vaccination. This intellectual ability demonstrates "cognitive empathy"— correctly inferring someone else's emotions, mental states (such as beliefs and intentions), and their reasons for experiencing them. In another instance, a doctor feels a patient's frustration and helplessness while discussing chronic pain. Such affect is termed "emotional empathy"—the ability to feel, in varying degrees of intensity, another's emotions. Finally, a doctor is moved to comfort a crying patient. This drive is "motivational empathy"—responding compassionately after perceiving another's distress.

Crucially, doctors do not need to experience emotional empathy in any given moment in order to convey motivational empathy. For example, embracing egalitarian moral principles,

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like the belief that everyone deserves dignified treatment, can motivate them to express empathy. So, too, even the utilitarian desire to meet professional goals, like using the expression of empathy to generate positive online reviews, could lead to the expression of motivational empathy, which in turn may lead to positive patient outcomes. Ideally, doctors would "always" feel the empathy they are expressing; however, when this is not possible in the moment, empathic skills can suffice. If, at the other extreme, a doctor *never* feels empathy, this could be a symptom of burnout and should be addressed right away. Nevertheless, even for those doctors who tend towards feeling less emotional empathy, it is possible for them to provide good, empathic care, and the healthcare system should still be structured to encourage them to do so.

The three types of empathy are interdependent, with the development of one enhancing another. By being open to the experience of emotional empathy, doctors can learn important information they might otherwise miss, such as accurately identifying how patients feel and want to be treated based on what these feelings convey. Knowing these things can enhance their cognitive empathy and positively inform how they display motivational empathy. For example, by being attuned to a patient's overwhelming fear of receiving a diagnosis, a doctor can sense the person wants emotional support and act accordingly. In turn, this can increase the likelihood that the patient will understand and adhere to the treatment plan. Conversely, if a doctor senses a patient only has mild concerns about test results, they can do what the situation calls for, perhaps calmly asking if there are follow-up questions.

Let us consider three ways that AI might improve the expression of medical empathy. First, imagine if AI could, in some meaningful sense "experience," or convincingly simulate a human doctor's cognitive, emotional, and motivational empathy. Then, we might see AI "doctors" appearing to have empathic conversations directly with patients, picking up on subtle cues, just like their human counterparts. Second, AI could function as a behind-the-scenes coach, helping human doctors fine-tune their bedside manner—a disposition that involves much more than delivering bad news—by suggesting appropriate words and phrases when doctors need (or think they might need) help. If this were possible, doctors might turn to AI to help them sound more caring. This approach might lead to patients receiving consistently compassionate messages, and the process could help doctors become more skilled at cognitive empathy (at least over time). After all, as doctors see, and accurately identify, more examples of empathetic communication, they might become better at recognizing what patients need and how they want to be talked to. Finally, AI could be used as an educational tool to simulate a patient's perspective. It could mimic a medical school role-playing exercise where a person (a student or actor) pretends to be a patient and provides post-interaction formative feedback in order to improve a doctor's cognitive and motivational empathy and communication skills.

Each approach has advocates. Arthur Garson Jr., a member of the National Academy of Medicine and a clinical professor of health systems and population health sciences at the University of Houston, has suggested a rather extreme option, urging us to prepare for a future when some human doctors are replaced with AI avatars [7]. Garson thinks it is possible, even likely, that someday AI-powered avatars displayed on computer screens could be programmed to look "exactly like a physician" and have "in-depth conversations" with "the patient and family" that are customized to provide "highly appropriate reactions" to a patient's moods and words. In this hypothetical scenario, the AI would be so advanced that it would be unlike anything that exists now.

Prominent AI experts and business leaders express similar views of the future. Mustafa Suleyman, the CEO of Microsoft AI, believes AI has already developed a remarkable capacity to simulate empathy [8]. Suleyman observes that "People said it [AI] would never be empathetic." And yet, he insists, "today, millions of people enjoy meaningful conversations with AIs, talking about their hopes and dreams and helping them work through difficult emotional challenges." Suleyman adds, "The greatest challenge of the 21st century is ensuring that "as we build out AI," we imbue it with "all that is special about our humanity," including "empathy" and "kindness." While acknowledging obstacles, this aspirational statement is an optimistic rallying cry for overcoming them.

A more moderate ambition for AI revolves around using large language models to improve real-time communication between doctors and patients. In this scenario, doctors could turn to generative AI to produce words, phrases, and sentences that they could then employ with a patient to express empathy. Anecdotal evidence suggests this a promising use of technology, such as Joshua Tamayo-Sarver's moving account of how ChatGPT saved the day in a California emergency department when he struggled to find the right words to connect with a patient's distraught family [9]. The AI helped him craft a sensitive explanation of why he chose a particular type of treatment that the family was initially worried about.

In a forthcoming book, *Move Slow and Upgrade*, the lead author, a philosophy professor, proposes the following possibility: considering integrating a version of generative AI into patient portals to help doctors sound more empathetic. Patients see portals as a lifeline, but doctors spend so much time fielding inbox messages that the correspondence contributes to their burnout [10]. Perhaps a win-win is possible. Doctors might improve patient satisfaction and reduce the number of follow-up questions patients ask by pushing an empathy button that proofreads, and perhaps even edits, their draft messages. In the meantime, doctors may even learn to express empathy better by being exposed to these AI-generated expressions.

A pilot initiative at the University of Rochester exemplifies the third educational scenario. One of us, the second author, is involved with the SOPHIE Project [11], a research group that has been studying whether doctors can learn to communicate more empathetically with human patients by interacting with a digital avatar that is powered by a large language model and designed to look and talk like a patient. SOPHIE uses a variety of factors (e.g., rate of speech, number of questions a doctor asked, and number of open-ended

questions) to provide personalized feedback in hopes of helping them improve their communication skills.

III. Creating A Foundation for Inclusive Ethical Discussion

Given the speed of technological change, now is the time to have thoughtful and inclusive conversations about the ethical issues at stake in the three possibilities highlighted above. Accordingly, we aim to establish a foundation for further critical and collaborative analysis. One of us is a philosopher, and the other is a physician, and we believe no single discipline or approach can provide all the answers. Beyond the wide range of credentialed experts, justice also requires that patients–who have their own expertise and are deeply impacted by their interactions with doctors–play a prominent role.

A. AI Empathy

At the heart of these discussions is a deep and vexing question: Will AI ever be able to experience the full range of human empathy? It might seem reasonable to say yes–and Garson's [7] scenario certainly presupposes the possibility–because humans are not the only empathetic creatures. Animals like chimpanzees appear to demonstrate it when they console "recent victims of aggression" [12]. and even octopuses express modifiable "prosocial behavior" [13]. But then again, AI is neither human nor animal, and it is hypothesized that the evolved biological bodies and social brains of pro-social primates make empathy possible [14].

According to the most skeptical scholarly perspective, AI will never have the rich human experience of empathy. Ultimately, we are, in crucial ways, fundamentally different from machines, and that difference cannot be bridged with more sophisticated programming. Carlos Montemayor, Jodi Halpern, and Abrol Fairweather contend that humans, unlike AI, can direct their attention and emotional receptivity toward someone else and, in that moment, imagine and, to some extent, sense "how it feels" for them to be in a particular "predicament" [15]. This direct empathetic connection draws on both conscious and unconscious processes, allowing us access to "second-person experience"—the point of view narrated in literature through the lens of the narrator referring to "you," not "I" (first-person) or "they" (third person). Crucially, with practice, taking the second-person perspective (a combination of cognitive and emotional empathy) can trigger a caring response (motivational empathy). This is because second-person experiences are more intimate than detached third-person observations, and they are less self-centered than thinking about what we would be feeling and thinking if we were in someone else's shoes. This seems to be intuitively true and fascinatingly may be expressed even at a basic physiologic level [16].

It is instructive to contrast this human experience with AI functionality. Montemayor, Halpern, and Fairweather [15] contend that AI systems cannot experience emotional empathy or interact with others with caring intentions because the technology is not conscious, lacks an affective life, and operates through rule-governed processes, such as cataloging and predicting information. To illustrate what this view on the fundamental limits of AI entails, imagine an AI-powered doctor avatar using facial recognition software when it consults with patients to detect information associated with body language. Given the hardware and software constraints, all it could do is infer what a hypothetical patient who fits a demographic profile might be thinking and feeling if an actual patient displayed classifiable facial characteristics, such as downcast eyes and lowered lip corners, after receiving a medical diagnosis. The AI would not feel anything when processing this information, and the distressed patient would not truly be seen as an individual. Hence, two concerning issues would arise.

First, since the AI could not relate to the patient as a unique individual, it might fail to detect important information. For example, a human doctor might only recognize the nuanced and contextual meaning of a patient's face after satisfying two conditions: getting to know the patient over the course of multiple visits and knowing firsthand what it is like to have related human experiences. Imagine a patient saying, "I'm only having one drink per week." At the same time, the doctor finds their attention drawn to conflicting body language and starts to wonder if the person is embarrassed to tell the truth.

Second, if the bot responded to the patient in a seemingly empathetic way, it would only be simulating empathy. A doctor bot that said, "I'm sorry to inform you," would be referring to feelings and intentions that it is not experiencing and, more fundamentally, cannot experience. The statement, therefore, would not only be inaccurate and insincere but also deceptive. It would create the false appearance that the bot both has the ability to care and actually does care at that moment. While some patients may feel good as the recipients of simulated care, the statement still would devalue the very idea of empathy. It suggests all that matters is whether potential recipients of empathy believe they are cared for, regardless of whether it is reasonable for them to hold such a belief.

To be sure, not all human doctors who sound empathetic truly feel that way in the moment. Some may, during any given patient encounter, simply go through the motions and do their best to make patients feel cared for. Jonathan Reisman, a physician, goes so far as to say, "It actually doesn't matter if doctors feel compassion or empathy toward patients; it only matters that they act like it [17]. In the same way, it doesn't matter that AI has no idea what we, or it, are even talking about" [17]. While this claim has some truth, it misses the mark in important ways. The fact that doctors do not need to feel the empathy they express in a particular moment does not justify the conclusion that it "doesn't matter" at all if a doctor never feels empathy towards a patient. The very possibility that human doctors can experience empathy—a possibility which, again, Montemayor, Halpern, and Fairweather [15] argue is off the table for AI–is a crucial factor that can help doctors and patients develop *justified trust*. Because a human doctor might very well mean what they say, patients are justified in believing their doctors might truly care about them. The possibility of genuine concern helps create the conditions for forming and maintaining a collaborative therapeutic partnership. When patients have good reasons to think that their doctors care about them, that conviction can help

create the conditions for forming and maintaining a collaborative therapeutic partnership.

To put this discussion of emotions in a broader context, professional environments influence what type of emotional labor people expect and why they consider it appropriate. If a co-worker smiles when you come into the office and says, "good morning," all things considered, it is expected that you respond in a way that conveys some politeness, even if you do not feel it, simply because workers can be expected to promote a cordial, professional environment. By contrast, when people are upset that a customer service representative gave them an insincere apology (with or without AI coaching), they are responding to a perceived ethical slight—to, for example, the person not caring, on a basic human level, about their time being wasted. Here, sincerity matters, but practically speaking, it can have low-stakes outcomes. However, when a patient feels ethically affronted by a doctor's insincerity, the stakes are higher: their dignity can be undermined, and the quality of their care may be compromised.

Insights from Jana Sedlakova and Manuel Traschel further illuminate why it is so easy for humans to misperceive bots as empathetic [18]. They rightly note that our species is easily confounded by chatbots that can have sophisticated natural language conversations with us. Our predictable misunderstanding of these highly unusual entities arises because the technology has core qualities that do not map onto our familiar way of understanding "tools" as fundamentally different from human "agents." On the one hand, chatbots can simulate being conscious individuals with rich inner lives that can form social relationships. On the other hand, and largely for reasons that Montemayor, Halpern, and Fairweather [15] emphasize, chatbots are not, and indeed *cannot* be, human "agents." The technology fundamentally lacks the necessary mental features of human agency (e.g., consciousness, autonomy, intentionality, a unified belief system, etc.), and even its impressive use of language occurs without understanding what words mean. Given these abilities and limitations, Sedlakova and Traschel [18] characterize chatbots as having a "hybrid nature," a strange type of existence that evokes familiar categories but does not correspond to them. Our lack of evolutionary experience with such confounding anthropomorphic beings makes it exceptionally difficult for humans to view chatbots in a clear-headed manner.

If Montemayor, Halpern, and Fairweather [15] are right, believing AI can adequately replace highly empathic doctors would be a grave mistake. Even if AI-powered doctor bots that simulate empathy can produce positive medical outcomes, their analysis suggests there may be compelling ethical reasons to avoid adopting technology, at least in some circumstances. If AI-powered doctor bots ever go mainstream, we do not know what impact the trend will have on the norms, values, and practices that today define medical professionalism. Indeed, it is hard to even predict the extent to which generational attitudes towards issues like dehumanization and deception will change.

Thinking critically about the range of possibilities requires a more in-depth discussion of potential conflicts between the medical ethics principle of beneficence and respect for persons beyond what Montemayor, Halpern, and Fairweather [15] provide. For example, more thought needs to be given to whether compelling practical or ethical reasons justify health systems declining to use a technically proficient AI doctor or patients opting out of consultation with one. It also requires extended discussions about healthcare disparities and fairness. For example, would using AI-powered doctor avatars exacerbate healthcare inequalities? Perhaps the wealthy will continue to have access to human doctors while underserved populations get saddled with second-rate digital services (or *vice versa*). Furthermore, since large language models have already generated outputs that "could potentially cause harm by perpetuating debunked, racist ideas," [19] we also should ask if the choices involved with creating the avatars reinforce negative stereotypes, such as suggesting certain races, genders, or ethnicities are more appropriately linked to being a doctor than others? There is also a pressing ethical question that we introduced earlier. Does engaging in critical discussions of AI and empathy detract attention from less speculative and risky approaches to addressing the empathy problem?

At the same time, we should be cautious of treating the analysis provided by Montemayor, Halpern, and Fairweather [15] as a definitive indictment of the limits of AI development. Many pressing questions remain unanswered. For example, given the potential for novel advancements in AIs, might Montemayor, Halpern, and Fairweather [15] may underestimate the possibility of AI developing some form of human-like empathy? Or might the evolution of AI go in a different direction, where it develops robust "robot empathy"—something different from human empathy but which, nevertheless, provides effective other-centered attunement? Ultimately, both in healthcare and more broadly, we should further consider whether AI empathy could ever be truly and sincerely empathy.

B. AI-Assisted Empathetic Communication

What about the second scenario, where doctors use AI to help them compose more empathetic-sounding messages to patients over medical portals? Imagine a doctor typing a response to a patient's question, hitting an "empathy button," and a large language model editing the text.

The first thing to consider is whether the technology will work. Large language models are prone to making mistakes, including confidently presenting inaccurate yet plausible information as if it were true. Still, much as concerns about hallucinations are valid, preliminary academic research, like a much-discussed article in *JAMA Internal Medicine*, suggests generative AI programs based on large language models can effectively simulate empathetic discourse [20].

More studies are needed, though. It is early days for learning the limits of generative AI. Also, researchers need to do more than gain additional data by repeating similar experiments. Ideally, they should find ethical ways to test the use of AI in real-world settings, not just in settings that serve as proxies for the real world, like having test subjects compare the

writing of human medical professionals and AI. Additionally, more research is needed into whether large language models provide inconsistent responses on matters that require empathy based on someone's identity and project empathy when people espouse harmful ideologies that should not be reinforced [21].

Of course, even if AI can, a good deal of the time, generate contextually appropriate, empathetic-sounding words and phrases, it will not always be accurate. Due to this limitation, a verification system will be necessary if doctors use AI. Perhaps one AI system could be used to proofread another. Or perhaps doctors will need to do the proofreading themselves. Either way, doctors have to be accountable for the messages they send, even if AI helps.

Designing a message-composing system with useful friction might be an effective way to promote accountability. For example, software could prompt doctors to answer comprehension questions to ensure they understand their messages. Or software might impose a time requirement for doctors to review messages before sending them. There are many possibilities to consider. Yet again, we need empirical testing.

Relatedly, time-management efficiency studies will need to be conducted. For example, will using AI in the way we have been describing not only benefit patients but also save doctors time? This might sound counterintuitive if doctors need to spend more time reviewing their notes. However, they may end up saving time in the long run. If patients feel better about their portal correspondence, they may have fewer follow-up questions.

Beyond the issues related to technical efficacy, there are a host of policy issues, from ensuring the technology is HIPPA compliant to being routinely audited and not used in ways that inadvertently harm vulnerable patients through personalized messages that display deceptive dark patterns [22]. But for the sake of argument, let us imagine, at least in principle, that all of these matters can be satisfactorily resolved. What ethical issues would remain?

For starters, it is important to note that a recent study suggests that while the *content* of an empathic message matters, so does the *messenger's identity* [23]. People have rated AI-generated empathic statements as better on average than human-generated ones if they did not know who or what wrote them. But the machine's advantage disappears once the recipient learns that a bot generated the words. Taking this response seriously raises questions about whether patients would believe, and would be justified in believing, that doctors who use AI to help them communicate are sincere. Are they using the technology to find the words they wish they could say if only they had more time to think of them or were not too burned out to compose? Or are they outsourcing the prose because, at bottom, they do not care? As with our discussion in the section above, the issue is not just whether patients trust their doctors but whether that trust is earned.

To this end, transparency is a central issue. Not only should doctors and patients alike have their autonomy respected by not being forced to use AI-assisted messages, but even if both parties agree, each should be clear about what they are consenting to. Embracing transparency, however, raises implementation questions. For example, how should AI-assisted messages be labeled? And is labeling alone sufficient?

Lastly, there are four remaining, especially difficult questions. What happens if the technology creates a multiple persona problem, where a doctor sounds like a saint online but is a robot in person? How can a new form of AI dependence be created to avoid further deterioration of human communication and skill—not just through automation bias, where doctors uncritically defer to AI systems, but potentially through a rewiring of their brains? What are the most appropriate safeguards to put in place to ensure that patients who want to try AI-mediated communication with their doctors but later decide to opt out can make a psychologically healthy transition as they "exit" the coached communication system? [24] And what steps can be taken to minimize the risk that patients who become accustomed to AI-mediated empathetic communication with their physicians online do not develop misplaced expectations for how people should talk to them in medical and non-medical settings where AI is not used?" [25]

C. AI-Assisted Training

Our third case is trying to improve a doctor's ability to communicate empathetically through a new, AI-driven approach to training where they talk with digital avatars that simulate patients and receive automated, evidence-based feedback on how well they converse. Again, one of us, the second author, is involved with this type of initiative: the SOPHIE Project at the University of Rochester. SOPHIE has the advantage of being designed to simulate the pedagogy created and refined by groups including Vital Talk [26], Ariadne Labs [27], Cleveland Clinic [28], and the University of Rochester's ACT Program [29], among others. Preliminary data from the pilot launch are promising, suggesting that SOPHIE improves communication skills, including the expression of empathy [30]. Consequently, larger clinical trials are currently underway.

Consequently, larger clinical trials are currently underway.

Using AI in this way has several advantages over more traditional means of training doctors, such as having them speak with human actors who role-play being patients. For starters, arranging for human actors to act like patients is far more logistically intense and financially expensive. The actors need to be paid, and their training can be costly. Doctors also have to be relieved of medical duty for at least half a day to attend an in-person course. Furthermore, physical space needs to be secured (and possibly rented), and food needs to be provided. Finally, due to all the costs involved, the training tends to be structured as a one-time, or at least relatively infrequent, event that offers little to no ongoing support to maintain and build skills. These limitations can impede long-term effectiveness.

In contrast, AI simulations of patients are much less expensive at scale. Additionally, doctors can be more flexible regarding where and when to interact with the AI since they can use the technology remotely and for comparatively shorter durations. Such flexibility can enhance long-term effectiveness because doctors will be able to interact with the AI—which

never gets tired and, barring technical issues, is always available—on a more ongoing basis. Doctors can even receive customized feedback for continual use and varied interactions tailored to the types of patients that doctors need the most help reaching. Finally, while human actors can make training feel more authentic, their non-standardizable performances raise issues of accuracy and reproducibility that may be less pronounced with AI.

However, even if AI-based communication training programs are developed that equal the efficacy and effectiveness of their human-based counterparts, two critical challenges remain. When compared to observed ratings, doctors tend to overestimate their communication skills, and so those most in need of training may be least likely to participate [31]. It is also unrealistic to expect that communication training, SOPHIE-like programs or otherwise, will be able to overcome system-level stresses and dysfunction, which are a major contributor to the empathy problem in the first place [32].

Finally, several ethical issues related to AI simulating patients need further exploration. Since patients may be wary of having a tool that can misunderstand their concerns, experiences, and needs represent "patient perspectives." Indeed, because human actors who role-play patients are authentically human and yet can also fall short in terms of portraying the desired aspects of a character, it is not clear what should be considered the "gold standard," and so further discussion is needed about how to assess human and AI patient personas and performances comparatively. Here is one of many questions that does not have a clear answer. Does it matter clinically or ethically that human actors can draw from personal experiences related to sickness and suffering? Relatedly, there is the issue of design justice regarding how to fairly and effectively incorporate different types of patients into the process of constructing AI avatars. For example, should interviews with patients be used as training data? If so, how can we ensure patients provide informed consent for such a novel use of technology—one they may have difficulty envisioning?

Additionally, given the misleading power of anthropomorphism, how should the avatars be depicted [33]? Should "they" have names or be referred to more generically? Should "they" introduce themselves as bots and perhaps refer back to their bot status throughout the conversation as a reminder of this fact? Should "they" physically resemble human patients or have less anthropomorphic appeal? And how should doctors be introduced to these bots? In addition to receiving preparative instructions, do doctors need a closing exercise to help them return to life outside of the simulation?

IV. Conclusion

Improving empathetic medical communication is daunting, but AI offers intriguing possibilities. To help focus medical ethics conversations on important issues, we have explored three novel scenarios: (1) AI "doctors" directly interacting with patients, (2) AI-assisted doctor communication in patient portals, and (3) AI-powered communication training that uses simulated patients. Each one represents a distinct approach to using AI, and our analysis reached the following conclusions.

First, we need to be cautious about falling for AI hype. While AI can replicate compelling, empathetic responses, the philosophical and practical questions concerning whether it will ever be able to truly experience genuine empathy remain unresolved. Overestimating AI's capabilities and undervaluing the difference between simulating empathy and authentically conveying it has profound implications for the doctor-patient relationship and the trust that should underlie it.

Second, there are compelling reasons to consider using generative AI to improve doctor-patient communication, but proper implementation will prove challenging. Above all, we believe the main goal of using generative AI should be to help doctors express the empathy they want to convey, not create a technological substitute for genuine human connection and care. While preliminary research on AI-generated messages that sound empathetic is encouraging, a good deal more inquiry is needed into actual patient responses to AI-mediated medical communication in real-world contexts. Furthermore, we only provided preliminary governance considerations for promoting the responsible use of generative AI in patient portals. A great deal of discussion still needs to take place to address the range of issues we identified, including the multiple persona problem, the risk of deskilling, and the complications surrounding accountability and transparency.

Third, AI-powered training simulations like the SOPHIE Project have many possible benefits, including providing doctors with more accessible, flexible, and standardized communication training. While preliminary results are encouraging, critical questions still remain about how to effectively design and administer these tools. We only scratched the surface of avatar representation and construction issues and the long-term consequences of supplementing traditional training with AI-powered ones.

Fourth, when AI tools are adopted in medicine, their use should not divert attention away from tackling the systemic issues that have, for far too long, been contributing to medicine's empathy problem. For all the benefits AI offers, fundamental workplace issues—from time pressures to administrative burdens and clinician burnout—should not be overlooked.

Ultimately, we have pursued a modest yet important goal. We have aimed to establish a foundation for thoughtful, interdisciplinary, multi-stakeholder conversations. We believe that the only viable path forward is for medical professionals, technologists, ethicists, patients, and policymakers to engage in an ongoing dialogue that addresses the benefits and risks of the issues we presented, identifies relevant issues that are missing from our analysis, and establishes appropriate protocols and parameters for ensuring AI can truly enhance medical care.

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empathy-problem-medicine/

BIOGRAPHIES

Evan Selinger is a Professor in the Department of Philosophy at the Rochester Institute of Technology. His research covers a range of issues in the philosophies of technology, especially privacy and ethics (including AI ethics). A prolific author, Prof. Selinger is currently working on a new co-authored book, co-authored with Albert Fox-Cahn, titled *Move Slow and Upgrade: The Power of Incremental Innovation* (Cambridge University Press). His previous book is Re-Engineering Humanity (co-authored with Brett Frischmann, featuring a Foreword by Nicholas Carr, and also published by Cambridge University Press). John Naughton selected *i* as one of The Observer's "Best Books of 2018" and one of "Thirty Books to Help Us Understand the World in 2020." Zadie Smith told *The Wall Street Journal* that it was one of her favorite reads of 2019.

Prof. Selinger regularly writes for newspapers, magazines, and blogs to foster critical conversations beyond the necessary yet narrow academic circles. He's currently a contributing writer at *The Boston Globe*. His essays and op-eds also appear in *The New York Times*, *The Washington Post*, *Wired*, *The Atlantic*, *Slate*, *The Wall Street Journal*, *The Nation*, *The Guardian*, *Salon*, *CNN*, *One Zero*, *Bloomberg Opinion*, *The Daily News*, *The Daily Beast*, *MTV News*, *Motherboard*, *The Christian Science Monitor*, *BBC Future*, *New Scientist*, *Aeon*, *Project Syndicate*, *Forbes*, and *HuffPost*. He's also written several review essays for *The LA Review of Books*.

To improve military policy, Prof. Evan Selinger is currently a member of the Institute for Defense Analysis's Ethical, Legal, and Social/Societal (ELSI) Working Group. We're contributing to DARPA-funded projects that use artificial intelligence. To enhance public policy, Prof. Selinger works with legal and advocacy organizations like the ACLU, Surveillance Technology Oversight Project, The Justice Collaborative Institute, and Fight for the Future. Prof. Selinger has also advised several companies, non-profits, and government organizations.

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ORIGINAL ARTICLE

In principle obstacles for empathic AI: why we can't replace human empathy in healthcare

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Abstract
What are the limits of the use of artificial intelligence (AI) in the relational aspects of medical and nursing care? There has been a lot of recent work and applications showing the promise and efficiency of AI in clinical medicine, both at the research and treatment levels. Many of the obstacles discussed in the literature are technical in character, regarding how to improve and optimize current practices in clinical medicine and also how to develop better data bases for optimal parameter adjustments and predictive algorithms. This paper argues that there are also in principle obstacles to the application of AI in clinical medicine and care where empathy is important, and that these problems cannot be solved with any of the technical and theoretical approaches that shape the current application of AI in specific areas of clinical medicine in which care for patients is fundamental. This is important, because it generates specific risks that may be overlooked otherwise, and it justifies the necessity of human monitoring and emotional intervention in clinical medicine. Consequently, difficult issues concerning moral and legal responsibility may ensue if these in principle problems are ignored.

Keywords Empathy · Consciousness · Attention · Care · Clinical medicine

1 Introduction

It is generally assumed in AI research that there are no "in principle" or a priori limitations about the applications and range of AI. In contrast, we argue that empathic AI is impossible, immoral, or both. Empathy is an in principle limit for AI. While the current argument is likely to generalize to other professions that rely on empathy, our attention here is confined to outcomes improved by empathy in clinical medicine and why AI cannot achieve these. Thus, our argument is not dependent on practical considerations and is limited in scope. Since it is an in principle problem, considerations about architecture, design, computer power or other pragmatic issues are not helpful in addressing it. But given that we consider primarily the area of patient care, rather than other aspects of medical applications of AI such as diagnostics, resource optimization or data gathering, in which AI has enormous potential for improving medical services, the difficulty we present does not amount to a categorically general objection to the use of AI in medicine.

The problem we present centers on the notion of empathy. As has been clarified recently in the literature,[1] various definitions of empathy in the literature emphasize three key components of empathy: (i) *emotional* empathy; (ii) *cognitive* empathy; and (iii) *motivational* empathy. Emotional and motivational empathy can be viscerally or biologically associated by *experiencing* emotions that lead to empathic concern for others, motivating us to offer help. Cognitive empathy, by contrast, is very different, because it allows us to *detect or recognize* the emotional mental states of conspecifics by representing their situation, thereby allowing us to identify salient interpretations of their emotions based on features of their expressions. Doing so can also lead to motivational empathy (offering help), but for quite different reasons, including manipulative ones. In fact, psychopathic patients are very good at cognitive empathy while lacking completely the remorse associated with emotional empathy. If we create AI that is very good at cognitive empathy but which is incapable of emotional empathy, are we creating "psychopathic" and potentially inhuman machines? This paper argues that this is a risk that must be examined and

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[1] Zaki (2017).

considered very seriously. We provide a more rigorous definition of these kinds of empathy by distinguishing two possible types of simulated empathy in AI applications.

Finally, while a comprehensive analysis of empathy and its relation to care may require a more thorough discussion of the nature of emotions and their relation to intentional action and rationality, the present discussion concerns the specific use of AI in medical care settings in which experienced empathy is essential. Therefore, a circumscribed and qualified approach suffices and in any case, delving into the philosophy of emotion and intentional action would lead us astray from our main goal. We focus on a classification of empathy in AI medical care situations to provide a clear basis for our argument, which is that in these settings, the lack of genuine or experienced empathy presents an in principle problem for the use of AI-simulated empathy, thereby creating risks in patient care that should be avoided for moral and legal reasons.

2 AI and experienced empathy

Having clarified the nature and scope of our argument, let us grant that AI will eventually simulate many forms of thought, skill, and cognition, including emotional reasoning. Like Moravec's paradox, most problems about AI development are conceived as hurdles that scientists will eventually solve. Some authors consider some challenges about moral machines as already addressed in systems that represent the satisfaction of goals without much autonomy, and as part of a larger AI project of developing more morally sensitive and autonomous AI.[2] Although some authors are more optimistic than others, and the range of issues about AI risk and value alignment are multiple,[3] there is consensus that there is no fundamental and insurmountable limitation for the development and application of AI in multiple fields, including morality. Here, we argue that there is at least one a priori limit in the application of AI: empathy, specifically in the field of medical care. Machine interactions in hospitals and similar care-based situations can achieve some neighboring phenomenon, but cannot, in principle, achieve empathy.

Clinical empathy, the use of empathy by nurses, doctors, therapists, etc. is emotion-guided imagining of what a particular moment, or slice of life (a time-slice or segment of one's life), feels like or means to the patient. The moment might be receiving a difficult cancer diagnosis. The slice of life may be going through a difficult post-surgical recovery. The clinician's ability to experience empathy in real time by resonating with a patient's emotional shifts while imagining what the situation is like from inside the patient's perspective (as-if in the subject position of the experience) enables not only more meaningful but more effective medical care for at least three reasons. Getting an accurate history is crucial for medical diagnosis. Did the patient first feel physically exhausted and then, when unable to work for a while, become depressed? Or did she feel exhausted every morning when thinking about depressing aspects of life and then lose her motivation to work? Replicated empirical studies show that patients disclose their histories selectively to physicians according to how emotionally attuned and empathic in real time they perceive their physician to be.[4] Studies show that they do not reveal information at first, but give emotional hints—saying that "my headache just kept coming back" with a lot of anxiety, till they sense that their physician is resonating with the importance of this moment in their story. When they sense this attunement, they reveal information, when they don't, they don't disclose.[5]

Second, effective medical care depends on patients adhering to treatment—the biggest cause of poor results in medicine (for people with access) is that approximately half of medical recommendations including prescriptions are not followed or taken as prescribed. The biggest predictor of adherence to treatment is trust in the physician, and it turns out that a major predictor (in some studies, the biggest predictor) of trust is the patient's perception that the physician is genuinely worried when they talk about something worrisome, that the physician is empathically accompanying them in real time.[6]

Third, a big part of medical care is helping patient's cope with bad news and regain agency to take necessary next steps to help themselves. Oncology patients who sense that their physician was empathizing with them when discussing their cancer diagnosis cope better, seeking treatment and support groups more actively than patients who did not feel so accompanied and had longer periods of confusion and anxiety after receiving difficult information.[7]

Note that all three of these benefits are based on the patient's experiencing the clinician's occurrent empathy in real time. They are not about whether the physician is highly knowledgeable about human behavior and good at predicting what certain people will do; while such knowledge and predictive abilities might have value in healthcare as well, and for that matter, might be exactly what AI systems can provide, this is not what was proven to improve outcomes in medical practice.

[2] Wallach and Allen (2009).

[3] Russell (2019).

[4] Eide et al. (2004); Finset and Ørnes (2017).

[5] Suchman et al. (1997); Halpern (2007).

[6] Roter et al. (1997); Kim et al. (2004); Halpern, "Empathy and Patient–Physician Conflicts, op.cit.

[7] Girgis and Sanson-Fisher (1995); Rosenzweig (2012).

These are virtues of attention that result in genuinely empathic interactions. Successful empathic communication manifests the skills and autonomy of agents—it is not accidentally the case that they empathize with each other; they empathize because they *consciously practice empathic attention*.[8] They know through their shared experiences what is salient in a situation and use their attention to virtuously select only what should be salient from an information set that contains a large amount of information that is not relevant to a specific situation. Moravec's paradox (1988) states that abstract and complex thought is easier to program than the easiest of motor skills. This has proven to be a fundamental challenge in robotics and AI in real, dynamic environments. This is not very surprising. Dexterous behavior has, after all, evolved through millennia of evolution, while the so-called "abstract thought" may be the result of the very recent ability humans have to express themselves linguistically. Here, we argue that the prospects for empathic AI face even greater challenges than Moravec saw for dexterous behavior. While relatives of empathy (compassion, sympathy) might be reproducible in AI systems and have some benefit in clinical settings, when empathy is properly understood it is clear that the capacities that manifest empathy are not capacities that any AI system can manifest.

However, our criticism is consistent with the distinctions mentioned above, and therefore, we are not arguing that AI cannot possibly achieve any cognitive capacities concerning empathy. In particular, AI may be quite good at cognitive empathy. It could accurately represent emotions (although this would be no trivial task) and properly relate situations with desired outcomes. However, because of the incapacity of AI to have emotional or experienced empathy, considerable risks regarding manipulation and unethical behavior need to be avoided, similar to the risks associated with psychopathological patients. Below, we introduce the notion of "empathy*" which could be considered a kind of cognitive empathy that is also insufficient for experienced empathy in medical care settings.

Turkle[9] has presented this problem as a slogan: simulated intelligence may be intelligence, but simulated emotion cannot be emotion. This is especially true of empathy. Humans, like other animals,[10] empathize with each other through the visceral and biologically based emotions our social brains evolved. Human empathy needs these emotional guides but builds upon them a cognitive apparatus that enables imagining not merely how the world "looks" from another person's perspective, but how the specific situation feels to another person inside it. Empathy is *imagining feeling* where the feeling is about something specific, that is the focus of another person's emotion.[11] What enables empathy includes emotional resonance with another's emotions, and the way that this resonance provides a kind of "mood" lighting/stage setting as well as a source of associative linkages that guide the activity of imagining another person's lived experience.[12]

3 Properly guided attention and empathy

It is because empathy has particularity—it is not a thought about how in general people feel, but an experience of imagining how this person feels in this particular situation, that the attention central to empathy is quite focused.[13] This emotion-guided cognition is a manifestation of our mental agency to direct our attention, which involves both conscious and unconscious components. We cannot just choose to resonate with another person, but we can choose to pay attention, try to imagine what they are going through, and in so doing will often find ourselves resonating, which improves our ability to imagine their situation, setting off a virtuous cycle. Importantly, we can have a second-order intention to be receptive that guides us to listen in this way to people over time, developing a habit of receptivity. Critically for this paper, the cognitive activities involved in not just paying attention but cultivating receptiveness to emotional resonance, which then enables imagining how their experience feels to them cannot be reduced to representations of rules and their application to cases.

Consciously empathic attention makes salient the needs of others, and most importantly, allows us to imagine having an experience that is filled in by the details of another person's life, not our own. That is, empathy is not imagining how we would feel if in their situation, but rather, how it feels to be in a situation delineated by their particular predicament.[14]

If genuine care needs to be provided to others in need, then consciously empathic attention must be involved. AI cannot provide consciously empathic attention, because empathy is based on our biological conscious and unconscious mental experiences and our attention capacities to select the most salient and important information for a patient in a situation of care. And this selection is rooted in biologic experiences like resonating with another's emotions. All AI will be able to do is represent the situation of

[8] Montemayor and Haladjian (2015); Haladjian and Montemayor (2016).

[9] Turkle (2005).

[10] de Waal (2019).

[11] Halpern (2001), pp. 85–92.

[12] ibid, pp. 92–95.

[13] Halpern (2014).

[14] Halpern, From Detached Concern to Empathy, op.cit., pp 82–85.

a hypothetical patient and apply it to a concrete data set concerning a specific patient according to some reliable algorithm or rule of inference. Therefore, AI cannot provide empathic attention, genuine care for humans—at best, they can provide emotionally unengaged care through representations and rules about cases.

The situation is much worse than the way Turkle put it. Simulated empathy is not only not really empathy; it is the opposite of empathy, because it is manipulative and misleading to the recipient. It generates responses in the receiver's social brain (i.e., the neural networks responsible for experienced and motivational empathy) that should not be triggered, because there is no biological agent in tune with their emotions at the other end. If so, we are doing a great disservice to those in need of genuine empathy and care by putting them in the "hands" of AI.

The unique elements of empathy that prove elusive to AI are grounded in Ludwig Wittgenstein's way of distinguishing the logical form of first person reports of occurrent psychological states from second or third person reports. For human beings, first person reports such as "My head aches" or "I expect him any moment now" are expressive and thus not in the business of depicting facts, or of being checked against the facts. We do not check our own winces or grimaces against the facts any more than we check "My head aches" against the facts. For the third person report that "Jones is sad" we cite behavioral evidence related to Jones to form the belief that Jones is sad. However, we do not do this for the first person case, we do not say "It seems to me that I am sad, for I am letting my head hang so." The upshot of this asymmetry is that we do not empathize with the psychological states of others through an analogical inference from our own case.[15]

According to Halpern, this is an important point in understanding the nature of empathy. Empathizing with another person is not a case of understanding what you would be experiencing were you in their circumstances, it is not analogically inferred.[16] If AI systems can only form analogically inferred presumptions about the psychological states of others, it cannot empathize with others. We empathize because we are implicitly guided by our own emotional experiences, including experiences of occurrent resonance, rather than as a consequence of an inference. Consider what such inference would have to be like: you observe the other's expressions from a third person perspective, match this to your internal state when expressing yourself in some similar way, then infer that this must be their internal state. Wittgenstein's point is that we do not match our external expressions of emotions to our internal states in this way. Empathy is much more automatic than this, because imputing subjective experiences to another is guided by occurrent resonance or similar emotional experiences rather than a third person process of inferential reasoning. Note for example that when we empathize with another, we have a sense that something is happening to us—we are having an experience—we experience empathy. This corresponds to being moved to imagine something rather than trying hard to imagine something. This is a problem in principle for AI, because empathy does not arise from understanding or observing one's own internal states to glean information that can then be matched to the other's external expressions, and this is the only way AI systems are capable of doing so.

4 Response to objections

Some may object, however, that this "in principle" problem is too unfairly anthropocentric. Why should AI be challenged with obscure notions concerning "what it is like" to experience empathy. Turing[17] called such objections "solipsistic", because there could be no way of verifying or sharing information. Thus, considerations about the qualitative character of subjective experience should not prevent us from designing AI that could enormously help in the field of medicine and other sectors where personal care is needed. Consider how advantageous it would be to have "caring" robots that attend to patients in the context of the current COVID-19 emergency. We already have some evidence that robotic pets that keep elders with mild-to-moderate dementia company, decrease loneliness and improve well-being[18] (real pets and even plants have provided the same benefits).[19] Could seemingly human companions or sites that convey empathy* do even more—help people feel understood as well as less lonely? This might bring the more powerful clinical effects that empathy, as opposed to company, can offer. Maybe, this is not genuinely felt empathy, but we can call it "empathy*." Empathy*, even if not biologically based, is empathic enough.

Again, there are logical, as well as ethical problems with empathy*. Empathy* is not a source of ethical and attentive care, because there is no affective resonance. However, there is a more fundamental problem with the argument in favor of empathy*, which requires the following assumptions: the conscious attention routines involved in empathy can be replaced, through deep-learning and predictive coding, with some kind of attention routine that is equivalent in results. Thus, it will produce behaviors that resemble attention-based

[15] Malcolm (1978).

[16] Halpern (2001), op, cit.

[17] Turing (1950).

[18] Portacolone et al. (2020).

[19] Stanley et al. (2014).

empathy by making salient the needs of patients through inferences or predictions about the patient's behavior.

The problem with this argument in favor of empathy* is that empathy* is not a form of attention either, and therefore, it cannot be based on an organizing helping attention that would enable such cognitive attunement through good inferences. This is also an a priori difficulty. If inference is guided towards helping others, the attention of an agent must be drawn from the premises of the inference to its conclusion. AI systems do not have the required autonomy, agency, and motivations to draw the conclusion from premises in a relevant and non-accidental way. Therefore, AI cannot be genuinely guided towards helping others through their agency, even if their results seem somehow equivalent to those of an empathic agent. AI lacks a helping intention towards another person as the basis of its attentional selection, because it does not have the appropriate motivational and inferential structure.

Mental action, motivations, and attention are hallmarks of solving problems concerning salience in a non-haphazard or accidental way.[20] We are building in too much luck and risk if we depend on AI, because it lacks empathy and empathy*. Since these are the only options, AI systems cannot provide empathic care. Autonomy and agency are the hallmarks of attentive agents—this is how their mental actions are not simply accidentally correct—and only attentive agents can care for others. The risk we create by ignoring this problem is quite serious: we put vulnerable people at the care of systems that, at best, can only accidentally "care" for them.

It might be objected that the argument above simply applies an old and well-known problem for AI in general to the particular case of empathy. It has been argued that conscious experience essentially possesses a phenomenal quality or "qualia." There is something that it is like for a subject to be in a mental state, and this is distinct from the information processing capacities of such states. Since AI cannot undergo qualitative mental states, AI cannot be in states of empathy. One might find this argument too quick and easy to merit much attention.[21]

Let's take a closer look at the role of qualia in empathy-based relations. While an empathizer might have various qualitative mental states, it is not essential that they have them, and attending to one's own mental states can even be counterproductive. The focus of the empathizer is the person being empathized with, it is an essentially *second person* perspective. While, the person receiving empathy may well need to be in qualitative states of mind to feel that they are being empathized with, the provider of empathy needs no qualitative states. Qualitative mental states are essential to the recipient of empathy, but not the giver of empathy. In fact, all that is required is for the interaction to involve emotional beings, which is compatible with judgment-based views of emotions.[22] Thus, our argument does not rely on the premise that AI would have to exhibit qualitative mental states to provide empathy.[23]

Halpern's account of 'affective resonance' may appear to require some qualitative states in both the giver and receiver of empathy. A subtle psychology of empathy will not preclude qualitative states in the empathizer, but will explain the nature of affective resonance without attributing any special qualitative states to the empathizer in the resonance achieved. And in fact, Halpern makes it clear that it is the empathizer's attentively following the flow of the receiver's emotional shifts that is crucial for the recipient of empathy to feel empathized with, and thus experience the benefits described above.[24] Such communicative attunement requires no special qualitative states in the empathizer. In practical terms, the improved clinical outcomes correlated with clinical empathy require that the recipient of empathy feel empathized with, so the role of specific qualitative affective states in the giver should be an open question. Rather than resonance with some unique qualitative feature of the care that empathy provides, the receiver might be said to resonate with the giver's *thoroughly second person, or receiver directed, attentional perspective.* The receiver is perceiving mental shifts in the giver of empathy, and this involves resonating with the thoroughly second person perspective of the giver, not a special and rather specific subjective state the giver is in. AI will have to replicate this to have clinically effective empathy.

The arguments above should suffice to establish the impossibility of empathic AI. However, there is a very different and equally problematic *ethical* obstacle. Even if the genuine article cannot be reproduced by AI, one can still defend an empathic AI program, because coming close to real empathy (e.g., as in empathy*) is good enough to put the attempt on solid footing. After all, human beings often have difficulty manifesting genuine empathy and we can only

[20] Wu (2011).

[21] For discussion, see Block (1980); Dennett (1990).

[22] Nussbaum (2001).

[23] Since our argument does not depend on phenomenal consciousness or qualia per se, but rather on the motivational aspects of empathic engagement through the second person/attentional perspective, it also constitutes an in principle objection to future artificially "conscious" systems that are conceived in terms of a non-organic substrate with a similar cognitive architecture. In particular, Susan Schneider's (2019) "substrate problem" or the difficulty that we don't know if chip-substrates with a complex architecture could be conscious or not, does not affect our argument, because these hypothetical systems would lack the motivational perspective that attentional engagement provides, which is an essentially social and emotional phenomenon.

[24] Halpern, op. cit., (2001). See also Murdoch (1971).

hope that, nonetheless, the effort is good enough to justify the attempt. On one hand, empathy* could be deployed towards people with mild-to-moderate dementia who may believe that it is real and thus experience clinical benefits. This would raise a classic ethical conflict between respect for persons and beneficence. One of us has argued elsewhere in the context of the robot pets that help reduce loneliness that when such results are based on deception, the violation of respect for persons is serious enough to shift to alternative means of decreasing loneliness such as supervised visits with real pets, or using AI to help people connect with real others.[25]

But what about when competent patients know that that their empathy* comes from AI and not a real person? Here, specific uses of something like AI empathy* may be ethically acceptable—for example, an AI responder akin to a smart journal to help people get feedback about how they are feeling (akin to biofeedback) that people can use to reflect. While we would not anticipate the same clinical benefits described above from this kind of intervention (though empirical research is needed to test that), there could be other benefits in self-awareness and perhaps deployment of cognitive behavioral therapy and other tools. Such interventions could enhance recipient's autonomy as well as well-being. The question that arises is why call the AI emotion "reflector" or "advisor" empathy*? Why not call it: AI feedback on emotions? Or something else that describes what it actually is? Most importantly, while getting feedback on one's own emotional states by a kind of smart journal may have real value, this is in fact not what has been shown to be deeply therapeutic about empathy when people are ill, suffering, grieving and afraid.[26] As described above, the therapeutic benefits of empathy, as opposed to journaling, meditating, and other salutary activities, have all been shown to depend specifically on the recipient's feeling that someone else is paying attention to them, curious to know more about them and worrying about them—in short caring about them. It misuses the term "empathy" to apply it to other important sources of understanding one's emotions. Given that health systems are under financial pressure to fund less human hours to provide the empathic care that patients need, we are concerned that labeling these AI programs "empathy" is a way to mislead the public (even if they know it's AI listening to them) and ultimately, to degrade expectations for human empathy in healthcare. Thus, we argue that it is ethically inappropriately to label even potentially very sophisticated AI emotional reflecting/advising as empathy.

In conclusion, empathic AI is thus either impossible or unethical. Impossible because of the lack of genuine empathy on the part of the AI. Unethical because of the risks empathy* produces not only for deception in specific instances, but for reducing and undermining the meaning and expectations for real empathy. The nature of the attention to another human being central to empathy is not exclusively a matter of rule following or good consequences represented under some algorithmic structure. Ignoring this could erode the normative expectation that when a person is suffering, they ought to elicit real human empathy, an expectation which is central to the morality of medical care as well as other relational practices.

[25] Portacolone et al. (2020), op. cit.

[26] Halpern (2017).

Declarations

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Abstract

Recently, there has been a large push for the use of artificial intelligence in medical settings. The promise of AI in medicine is considerable, but its moral implications are insufficiently examined. If AI is used in medical diagnosis and treatment, it may pose a substantial problem for informed consent. The short version of the problem is this: medical AI will likely surpass human doctors in accuracy, meaning that patients have a prudential reason to prefer treatment from an AI. However, given the black box problem, medical AI cannot explain to patients how it makes decisions, yet such an explanation seems to be required by informed consent. Thus, it seems that doing what is best for patients (treatment via AI), even if patients want to permit this, might be prohibited by medicine's commitment to informed consent. Conflicts between beneficence and autonomy are not new, but medical AI poses a novel version of this conflict, because this problem is one in which even if the patient says they want to use their autonomy to receive better care, the commitment to autonomy (via informed consent) seems to block them from doing so. Given this dilemma, should we abandon informed consent, or should we not use medical AI? My thesis is that we can have our cake and eat it too; we can use opaque AI in clinical medicine and retain our commitment to informed consent, although it may require revising our understanding of informed consent.

Keywords

Consent; Information; AI; Explainability; Medical AI

Does Black Box AI In Medicine Compromise Informed Consent?

Recently, there has been a large push for the use of artificial intelligence in medical settings. The promise of AI in medicine is considerable, but its moral implications are insufficiently examined. If AI is used in medical diagnosis and treatment, it may pose a substantial problem for informed consent. The short version of the problem is this: medical AI will likely surpass human doctors in accuracy, meaning that patients have a prudential reason to prefer treatment from an AI. However, given the black box problem, medical AI cannot explain to patients how it makes decisions, yet such an explanation seems to be required by informed consent. Thus, it seems that doing what is best for patients (treatment via AI), even if patients want to permit this, might be prohibited by medicine's commitment to informed consent. Conflicts between beneficence and autonomy are not new, but medical AI poses a novel version of this conflict, because this problem is one in which even if the patient says they want to use their autonomy to receive better care, the commitment to autonomy (via informed consent) seems to block them from doing so. Given this dilemma, should we abandon informed consent, or should we not use medical AI? My thesis is that we can have our cake and eat it too; we can use opaque AI in clinical medicine and retain our commitment to informed consent, although it may require revising our understanding of informed consent. To clarify, I intend this paper to be about the practice of using AI to diagnose patients and suggest treatment plans, not the practice of using AI to actually perform surgery.

This paper will proceed in the following order. In section I, I explain in more detail the promise of medical AI. In section II, I show that black box AI in medicine is incompatible with our current understanding of informed consent. In section III, I consider and reject explainable AI as a solution to this problem. In section IV, I advance a solution to this problem in the form a revision to our

understanding of informed consent (which I call higher-order consent) that allows us to maintain our commitment to informed consent *and* use AI in clinical settings.

I. The Promise of Medical AI

Although much of it is still nascent, the potential uses of AI in medicine are substantial. A review article in *Nature* identified numerous potential uses for AI in clinical practice, including "disease diagnosis, interpretation of patient genomes, treatment selection, automated surgery, patient monitoring, [and] patient risk stratification for primary prevention" (Yu et al. 720). My specific interest in this paper concerns the use of AI in diagnosis and treatment recommendation in clinical medicine.

Within diagnosis and treatment, AI has many applications. To name a few, AI can be used to predict and prevent hypoxaemia, which is "an unwanted physiological condition known to cause serious patient harm during general anaesthesia and surgery" and "is associated with cardiac arrest, cardiac arrhythmias, post-operative infections and wound healing impairments, decreased cognitive function and delirium, and cerebral ischaemia" (Lundberg et al., 749). Lundberg et al. report that machine learning algorithms can enable anesthesiologists to predict hypoxaemia twice as accurately:

> The [AI] system, which was trained on minute-by-minute data from the electronic medical records of over 50,000 surgeries, improved the performance of anesthesiologists by providing interpretable hypoxaemia risks and contributing factors...Our results suggest that if anesthesiologists currently anticipate 15% of hypoxaemia events, with the assistance of this system they could anticipate 30%, a large portion of which may benefit from early intervention (749).

AI can also be used to diagnose diabetic retinopathy. The FDA approved an AI for this purpose, which has been in use at the University of Iowa (He et al., 34). Abràmoff et al. report that this AI (with 819 patients) was 87.2% sensitive (meaning able to correctly identify cases where the person does not have the condition) and 90.7% specific (meaning able to identify cases where the person does have the condition), while "previous studies have shown that board-certified ophthalmologists that perform indirect ophthalmoscopy achieve an average sensitivity of 33%, 34%, or 73% compared to the same ETDRS standard" (Abràmoff et al., 3). This is a stunning difference in abilities between human doctors and medical AI, one which both inspires confidence in the abilities of AI and some skepticism about the abilities of human doctors (in this domain). Abràmoff et al. describe this as the "ability to bring specialty-level diagnostics to primary care settings," with the possibility "to help prevent vision loss in thousands of people with diabetes annually" (1).

More generally, AI has huge implications for any area of medicine that relies on imaging to make diagnoses, which includes "radiology, ophthalmology, dermatology and pathology" (Yu et al., 722). In dermatology, "convolutional neural networks trained on 129,450 clinical images achieved dermatologist-level accuracy in diagnosing skin malignancy" (Yu et al., 723). Furthermore, "the deep-learning algorithm outperformed the average dermatologist in a comparison of the algorithm predictions and the assessments by 21 dermatologists on a set of photographic and dermoscopic images" (Yu et al. 723).[1] Some in the literature have gone as far as to suggest eliminating entire specialties, such as Hinton's suggestion that AI renders radiologists obsolete. In 2016, Hinton boldly claimed that "people should stop training radiologists now. It's just completely obvious that within 5 years, deep learning

[1] Here, Yu, Beam and Kohane are referring to Esteva et al. (2017).

is going to do better than radiologists." Additionally, Chockley and Emanuel have said that "the ultimate threat to radiology—the one that could actually end radiology as a thriving specialty—is machine learning" (1415).[2]

The motivation for using medical AI is that it (among many other things) is cost cutting and produces better patient outcomes. If we replace certain features of clinical medicine with AI, we can eliminate the costly labor of technicians, hopefully reducing the cost of medicine. Additionally, (and more important), AI can produce better medical outcomes for patients than human doctors can. As Yu, et al. put it, "AI systems have specialist-level performance in many diagnostic tasks...[and] can better predict patient prognosis than clinicians" (722).

I'll remain neutral on what the future of AI medicine would look like in the clinic. It could be just AI, AI and humans, etc. As will become clear, my argument is contingent on a future scenario in which it is very likely that treatment via human doctor will remain an option for all who object to treatment via AI.

II. The Problem for Informed Consent

In this section, I argue that medical AI clashes with current doctrines of informed consent. For the sake of argument, I assume that (at some point) medical AI will surpass human doctors in many fields and will be reliably able to diagnose and suggest treatments that produce better medical outcomes for patients than human doctors could. While this is a large assumption, I make it because I want to determine if *AI itself* poses a problem for informed consent, not if the potential inaccuracy of AI poses a problem. As I will set up the problem, *if* opaque AI really is better for patients than human doctors, we need to confront how AI being a black box interferes with consent.[3] If it turns out that AI in medicine isn't actually better for patients, then we don't need to ask the question about consent. My suspicion is that most consent-based objections to AI in medicine really reduce to a worry that AI is not medically better for patients, not from an actual conflict with informed consent. Another issue I will bracket is how an AI would incorporate patient input into the process of shared decision making. For the sake of the argument, I'm assuming that medical AI will have the ability to incorporate a patient's preferences and beliefs into its treatment recommendation.[4]

To be abundantly clear, I am not claiming that medical AI is currently better than human doctors or that it will be at any near point in the future. I want to investigate the ethical implications of the possible future world in which AI does surpass doctors. I bring this up, because it is tempting to reply that medical AI isn't or won't soon be better than human doctors. That may be true, but it would

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[2] It is important to note that there are those who argue that medical AI is not as impressive as it has been claimed to be. For example, a 2019 meta-analysis "found the diagnostic performance of deep learning models to be equivalent to that of health-care professionals. However, a major finding of the review is that few studies presented externally validated results or compared the performance of deep learning models and health-care professionals using the same sample. Additionally, poor reporting is prevalent in deep learning studies, which limits reliable interpretation of the reported diagnostic accuracy" (Liu et al., e271).

[3] Gray Grant, Behrends, and Basl defend explainability (not just in medical AI) on the grounds that opaque AI is not always more accurate, in part because it violates procedural duties that make it likely to ignore relevant evidence and to include evidence that ought to not be included.

[4] See Holm for a discussion of shared decision making and medical AI. Recent work on using AI to predict the preferences of incompetent patients may also be relevant here; see (Earp et al. 2024).

not affect my argument unless there is compelling reason to think that it's extremely unlikely that AI will surpass humans in medicine.

One might wonder how issues of algorithmic bias and fairness relate to this issue. Given that the question of diagnosis and treatment is not, necessarily, a question of distributing resources across competing participants, fairness questions are less relevant to this issue. However, issues of bias are extremely relevant. An algorithm may be biased in a way that skews its recommendations along racial lines. For example, if an algorithm thinks that African Americans live shorter lives on average or have higher pain tolerance than other races, it might suggest treatments lead to worse medical outcomes for African Americans. This is an extremely important problem. However, it is beyond the scope of this paper to propose and account of how AI can avoid biased data. For the sake of argument, I assume that there will eventually be a way to eliminate data from algorithms that might lead to racially unequal outcomes. Additionally, my question concerns if even a well-functioning, non-biased but opaque algorithm would violate consent.[5]

If it's true that medical AI will produce better outcomes for patients, then it seems that informed consent poses a barrier to the best outcome for the patient. Informed consent must be informed. But, given the black box nature of AI, doctors would be unable to give the patient an explanation of why the AI made a particular diagnosis and treatment recommendation. Given this, the patient's right to an explanation is frustrated, she cannot make an informed decision, and thus she cannot validly consent. So goes the problem. As Schiff and Borenstein put it:

> Relevant information includes the purpose of the treatment, its potential benefits and risks, and possible alternative treatment options. Yet the novelty and technical sophistication of an AI device places additional demands on the informed consent process. When an AI device is used, the presentation of information can be complicated by possible patient and physician fears, overconfidence, or confusion. Moreover, for an informed consent process to proceed appropriately, it requires physicians to be sufficiently knowledgeable to explain to patients how an AI device works, which is rendered difficult by the black-box problem...The result can be a system largely unintelligible to humans beyond its most basic inputs and outputs (139-140).

I will address the nature of the black box problem more in the next section when explainability comes up; for now, it will suffice to say that when black box AI is involved, patients are cut off from receiving an adequate explanation (according to current standards) of how their treatment decision was justified. Without this explanation, their consent is not sufficiently informed and thus not valid.

One might wonder if patients in this scenario (who know that treatment via AI is likely better for them) could waive their right to an explanation. This would be a way to say that the patient gave valid consent to the AI treatment plan. However, current conceptions of informed consent in medicine would not allow this, as we require that patients are at least able to receive an explanation of their diagnosis and treatment. We may allow patients to waive their rights to an explanation of their treatment, but we presently require that they have the possibility of accessing this right. In other words, our current understanding of informed consent would allow a patient to waive their right to an explanation (although some question even that), but this is only in a context where this patient

[5] For recent discussions of algorithmic bias and fairness, see Milano and Prunkl (2024), Di Bello and Gong (2023), Babic and Johnson King (2023).

could, if they desired, receive an explanation. When it comes to opaque AI, the problem is not whether patients can waive a right to an explanation that they otherwise could easily access; the question is whether consent can be informed in a case where patients could not access explanation of their diagnosis and treatment plan even if they wanted to. It is not all that controversial to suggest that, sometimes, patients may autonomously waive their right to an explanation.[6] However, that occurs in a context where a patient *could* receive an explanation if they so desired. In the context of opaque AI, no such explanation can be given, even if the patient wants one.

The previous paragraphs may have dealt too quickly with what bioethicists call the *right to not know*.[7] Although it often comes up in the context of genetic testing, the basic idea can generalize to any medical context. The idea is that patients have a right to not know certain pieces of medical information that they select to not know, for whatever reason. This fits into the dialectic, because it may be a way to say that opaque AI does not actually conflict with current practices surrounding informed consent. If patients can already waive their right to be informed, how is treatment via opaque AI different from this, as they could simply waive their right to be informed in this case as well?

The debate on the right to not know is extensive, with some arguing that patients lack this right altogether, while others argue that patients do have this right, so long as exercising it won't cause harm to third parties.[8] However, the right to not know debate comes in a context where it is a given that the patient can know the information if they want to. As mentioned in the previous paragraph, when it comes to opaque AI, this is not an option at all. In that manner, appealing to the right to not know will not make treatment via opaque AI compatible with our current standards of informed consent.

Several have argued that black box AI is incompatible with informed consent and thus that we should prohibit black box AI. Kundu claims that "relying on devices whose logic is opaque violates principles of medical ethics...Patients' autonomy and informed consent may be jeopardized if they are unable to question the AI system" (1328). In a related vein, Maclure argues that explainable AI in medicine is a requirement of Rawlsian views of public reason; as she says,

> Regulators ought to impose an explainability requirement upon the (public and private) organizations that choose to delegate decision-making to AI programs when the wellbeing, rights, and opportunities of citizens are at play. For example, if IBM's Watson Health makes a surprising diagnosis or recommends an unconventional treatment, the medical team has the duty to explain to the patient why the diagnosis was made or the treatment recommended. This is required by the norm of informed consent (12).[9]

Wadden argues that opaque AI clashes with standard conditions for informed consent:

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[6] As Eyal says, "what we may call 'waiveable waivers' of informed consent rights ('Please spare me the full disclosure or decide in my name unless I ask to resume full disclosure or active control') seem relatively easy to accept and fairly consistent with autonomy."

[7] The debate on the right to not know begins with Ost's 1984 article.

[8] Adorno defends the latter position, namely that there is a right to not know but that it is not absolute and is violable when doing so would protect third parties. Others (e.g. Hull) in addition to Ost have argued that patients lack the right to not know.

[9] See Da Silva (2023) for a response to Maclure.

> Opaque systems...pose a particular problem for disclosure and understanding, both of which are part of the informational components of informed consent. Indeed, opacity makes it difficult for a healthcare professional to disclose why a system makes a decision, and it stands in the way of a patient understanding why the system makes its decision. A patient can reasonably be deemed competent enough to make a decision and able to voluntarily make a choice, but without proper information they cannot provide informed consent (decide and authorize). At best, in these circumstances, we might be able to claim the patient is assenting, but this does not hold the same legal or procedural weight as informed consent (96).[10]

Watson et al. endorse a similar line of argument:

> One frequently cited obstacle to machine learning's wider clinical adoption is a lack of understanding among patients and doctors about how predictions are made. This is especially true of some top performing algorithms, like the deep neural networks used in image recognition software. These models may reliably discriminate between malignant and benign tumours, but they offer no explanation for their judgments...If doctors do not understand why the algorithm made a diagnosis, then why should patients trust the recommended course of treatment? Is informed consent even possible without some grasp of how the model reached its conclusion? (2)

In the context of patient-centered medicine, Christian Bjerring and Busch argue that:

> In striving to establish a state of shared mind between patient and practitioner, exchange of information, explanation, and understanding become key components...But when we include black-box AI systems into the mix, it becomes potentially impossible to uphold this practice. The core reason is simple: since black-box AI systems do not reveal to practitioners how or why they reach the recommendations that they do, then neither can practitioners who rely on these black-box systems in decision-making—assuming that they honor their epistemic obligation—explain to patients how and why they give the recommendations that they do. Yet, for patients to make decisions in autonomous and rational ways, it is a requirement that they have "the capacity to make sense of the [medical] information presented and can process it rationally to reach a decision that furthers their health care goals" (Bernat and Peterson 2006, p. 88).[11]

To sum this section up, patients have a right to an explanation, and opaque AI seems incapable of giving such an explanation. And, even if patients have a right to not know the information that would be provided in an explanation, opaque AI presents an importantly different case in which patients can't know this information even if they wanted to. For these reasons, opaque AI conflicts with our current practices surrounding consent.[12] This means that pursuing what, by stipulation, is best for patients would require medical interventions that lack the valid consent of patients.

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[10] Wadden ultimately defends a "grey box" solution, which involves AI systems that "include some combination of opacity and transparency" (2021, 94).

[11] Christian Bjerring and Busch 360-361. The authors are quoting from Bernat and Peterson 2006.

[12] This version of the problem differs from the problem addressed by Cohen; Cohen is concerned with whether patients have a right to know that AI was used in their diagnosis (1441). My question is, even if patients know that AI was used, can they have sufficient understanding of its recommendations to give informed consent.

III. Explainability To The Rescue?

As it stands, it seems that autonomy (via informed consent) is winning at the expense of beneficence (via the use of medical AI). As an anti-paternalist, I am committed to autonomy winning in a *genuine* conflict between autonomy and beneficence. Ideally, we can navigate the tension in such a way that there is no genuine conflict.

An initially tempting way to dissolve the seeming conflict between autonomy and beneficence is explainable AI. Like all terms in AI discourse, "explainability" can mean many different things; I am using the term to mean "a characteristic of an AI driven system allowing a person to reconstruct why a certain AI came up with the presented predictions" (Amman et al., 2).[13] Along similar lines, He et al. describe transparent AI as meeting the requirement that "humans should be able to understand or interpret how a given technology reached a certain decision or prediction" (He et al., 31). In other words, an explainable medical AI would be able to explain, to some degree sufficient to meet the standard of explanation for the patient, how and why it reached its diagnosis and treatment recommendation. If it were truly the case that all AI could be sufficiently explainable in clinical terms that could be intelligible to both doctors and patients, this would clearly be the best system and would pose no problem for informed consent. However, there are reasons to think that this is not really a solution to the problem.

It's helpful to distinguish between two different kinds of opaqueness (the opposite of explainability) in AI: (1) *in-practice opaqueness* and (2) *in-principle opaqueness*.[14] If a system is in-practice opaque, this means that we could explain how it makes decisions if we had unlimited time and computational power; however, given the realities of the clinic and the fact that doctors are not computer scientists, these AI will almost always not be explainable to patients. Yet, they *could* be explained. If a system is in-principle opaque, this means that even with unlimited time and computational power, we *could not* understand why it reached its decision. The claim 'opaque medical AI is incompatible with informed consent' could be understood on either view of opaqueness. In-principle opaqueness (by definition) rules out explainability, so only in-practice opaque AI systems would allow patients to receive an explanation and thus be a potential solution to our problem. The question is whether explainable/in-practice opaque AI would solve the problem by allowing us to use AI in medicine while fully respecting informed consent. Unfortunately, this approach does not solve the problem.

It's likely that a medical AI that is explainable will be less accurate than an in-principle opaque AI. This means that explainability comes at the cost of high performance and accuracy, ultimately producing a worse medical outcome for patients. This claim is echoed by many working on the

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[13] Amman et al. note that "explainability has many facets and, unfortunately, the terminology of explainability is not well defined. Other terms such as interpretability and/or transparency are often used synonymously" (2).

[14] As Nicholson Price explains: "algorithms can be opaque for multiple reasons. Sometimes, algorithms are nontransparent because, while they may rely on explicit rules, those rules are too complex for us to explicitly understand—for example, patients whose measurements place them in a particular region of n-dimensional (where n is large) characteristic-space are at a higher risk of stroke. In particular, these rules may be impossible to explain or to understand by following the process of scientific/medical discovery: mechanistic lab experiments followed by confirmatory clinical trials. Other times, the relationships used in a blackbox algorithm are literally unknowable because of the machine-learning techniques employed—that is, no one, not even those who programmed the machine-learning process, knows exactly what factors go into the ultimate decisions. A key distinguishing feature of black-box algorithms, as the term is used here, is that it refers to algorithms that are inherently black box (i.e., their developers cannot share the details of how the algorithm works in practice)—rather than to algorithms that are deliberately black box (i.e., their developers will not share the details of how the algorithm works)" (430).

technical side of AI. As He et al. say, "in certain instances, enforcing transparency and interpretability can potentially result in decreased accuracy or predictive performance of a model (2019, 31). Ursin, Timmerman, and Steger add that "there is a tradeoff between accuracy and explicability: the more explicable an AI system is, the less accurately it performs (144). As Babic et al. say, "especially in cases with massively high dimensionality—such as image recognition or genetic sequence analysis— limiting oneself to algorithms that can be explained sufficiently well may unduly limit model complexity and undermine accuracy" (286). In short, explainable AI will be worse for patients than in-principle opaque AI would be.

For explainable AI to solve the problem, it would have to get us *both* autonomy and beneficence, but it seems that explainable AI merely presents another way of preferring autonomy *over* beneficence. Suppose that explainability (for the time being or forever) comes at the cost of accuracy and ultimately leads to worse outcomes for patients. If that is the case, patients themselves have a prudential reason to prefer in-principle opaque models over explainable models. Unless patients want a treatment that is less likely to help them than another available option, they should prefer the in-principle opaque AI. Of course, explainable AI may still be more accurate than a human doctor. But, the point stands that patients have a prudential reason to prefer the *best* medical advice. Again, if explainable and highly accurate AI were possible, it would clearly be the optimal option. The conflict between informed consent and medical AI would entirely dissolve if this were an option. What I'm suggesting is that absent that, we can still get the benefits of in-principle opaque AI in medicine even if we cannot achieve high functioning and explainable AI. My argument for this claim comes in the next section.

IV. Higher-Order Informed Consent As A Solution

In the previous section, I argued that explainability, if possible, will come at the cost of better patient outcomes and is thus not a solution to our problem. Ideally, a solution to the problem would achieve both a commitment to informed consent and a commitment to beneficence. Here, I'll argue that even if we are faced with either kind of opaqueness (in-practice or in-principle) in medical AI, patients can autonomously make the decision to receive treatment and diagnosis from such an AI.

So, where does informed consent fit? I defend the view that we can get the diagnostic benefits of AI while maintaining a commitment to informed consent if we allow for a kind of *higher-order informed consent*. By this, I mean a kind of consent where patients may not be informed about the first-order details of how the AI reached its diagnosis and treatment plan but are informed of higher-order information about the reliability of the AI itself.

In discussions of informed consent, bioethicists have gone to great lengths to specify what information patients must know to count as being sufficiently informed. However, the philosophical discussion of informed consent has thus far not distinguished between two different kinds of information that might be relevant for a patient. What I will call *first-order medical information* is information about the doctor's process of clinical reasoning. For example, "your white blood cell count is elevated, which indicates condition X, which I will prescribe drug Y to solve, because drug Y has been shown to alleviate condition X because of reasons A, B, and C." Compare this to what I will call *higher-order medical information*, which is information about the doctor themself or the treatment itself. For example, a doctor saying only the following, without an explanation as to why or any discussion of causal mechanisms would count as higher-order information: "in all the cases of X I've treated before, 99% have gone on to live healthy lives" or "drug Y is likely to cure you, but we don't have a

causal explanation for why this is the case." In canonical treatments of informed consent, this distinction between higher-order and first-order medical information is absent.[15] Consider the following authoritative documents that pertain to consent:

> **The Nuremburg Codes:** the person involved...should have sufficient knowledge and comprehension of the elements of the subject matter involved, as to enable him to make an understanding and enlightened decision. This latter element requires that, before the acceptance of an affirmative decision by the experimental subject, there should be made known to him the nature, duration, and purpose of the experiment; the method and means by which it is to be conducted; all inconveniences and hazards reasonably to be expected; and the effects upon his health or person, which may possibly come from his participation in the experiment (1448).

> **The Declaration of Helsinki:** In medical research involving human subjects capable of giving informed consent, each potential subject must be adequately informed of the aims, methods, sources of funding, any possible conflicts of interest, institutional affiliations of the researcher, the anticipated benefits and potential risks of the study and the discomfort it may entail, post-study provisions and any other relevant aspects of the study. The potential subject must be informed of the right to refuse to participate in the study or to withdraw consent to participate at any time without reprisal. Special attention should be given to the specific information needs of individual potential subjects as well as to the methods used to deliver the information.[16]

In their extremely influential textbook, Beauchamp and Childress say that:

> Professionals are generally obligated to disclose a core set of information, including (1) those facts or descriptions that patients or subjects usually consider material in deciding whether to refuse or consent to the proposed intervention or research, (2) information the professional believes to be material [they go on to list 3 more conditions not relevant for the question at hand].

These sources are extremely influential in the history of informed consent, and none of them distinguish between which level of information must be given to patients. Additionally, although it's too long to reprint here, the Council for International Organizations of Medical Sciences lists numerous items research participants must be informed about to give valid consent. Nowhere on their list is any distinction made about levels of information.[17] Additionally, the American legal system seems to focus on first-order information and to not require higher-order information; as Cohen says, "The substantial majority of courts [in the US] have rejected the notion that the failure to disclose the physician's experience or qualification breaches the duty of informed consent, on the theory that only information about the procedure itself is material" (1435).

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[15] Although proving a negative is difficult, in my career researching informed consent, I've never seen anyone invoke this distinction.

[16] Perhaps it's not fair to use standards of consent for research ethics to apply to clinical medicine. I disagree; the standard of consent should not change (or at least should not be more relaxed) in clinical medicine as opposed to research trials.

[17] See Council for International Organizations of Medical Sciences (103-106).

Despite the distinction between higher-order and first-order medical information being absent from various philosophical and policy statements concerning what information is required for consent, it seems clear to me that the implicit understanding of informed consent among bioethicists is that, when possible, patients must be informed of the first-order information. In other words, bioethicists would likely regard access to first-order medical information as a necessary condition for consent being sufficiently informed. Clearly, the model of the patient as the recipient of information which they use to make an autonomous and informed choice is in tension with a world in which patients are only being told higher-order information. The aspiration of the informed consent movement was to depart from a model of the doctor-patient relationship in which patients deferred to the doctor and to move instead to a model where the patient uses the clinical information given by the doctor to make an informed decision about their care. A medical system in which doctors only tell patients higher-order medical information (with no chance for patients to ask "why" questions) would much more resemble the kind of "trust me, I know what's best" view that the informed consent movement is a response to. Not to belabor the point, but it bears repeating. Although bioethicists have not made the distinction between higher-order and first-order medical information, it seems very clear to me that they regard first-order information as being required for informed consent. If the requirement of informed consent were to permit higher-order information to count as sufficient for informed consent, then patients could be sufficiently informed even if they were only told "I don't know the details of why the treatment works, but I know that it works in 99% of similar cases." Proponents of informed consent as it is currently practiced would not regard this as sufficient information, indicating that they instead view first-order information as necessary for informed consent.

In our current understanding, it would not be enough if the patient knows that the doctor is very reliable but knew nothing about the reasoning in support of the treatment. Higher-order consent, as I've proposed it, violates this view by suggesting that patients don't need to be informed about the first-order reasoning of their doctors, so long as they are informed about the higher-order reliability of their doctors.

Against this backdrop, my proposal is that if we allow higher-order medical information to be sufficient for consent to be informed, then patients can still be sufficiently informed even if doctors use in-principle opaque medical AI. In a world where medical AI is widely used, we would have access to higher-order medical information about how accurate and reliable an algorithm is, and if this information is sufficient for consent, then patients can consent to being treated by opaque AI.[18] We can express this view as the following thesis:

> **Higher-Order Consent:** if patients are only informed of higher-order medical information about their treatment/diagnosis (and cannot access any first-order medical information about their treatment/diagnosis), they can still give valid, informed consent.

Compare this with the view that I take it that most bioethicists (implicitly) endorse:

> **First-Order Consent:** valid, informed consent requires that patients have the option to receive an explanation of their treatment/diagnosis worded in first-order medical information.

[18] This would be in concert with Kiener's claim that patients should be informed "that even the best AI comes with the risks of cyberattacks, systematic bias, and a particular type of mismatch between AI's implicit assumptions and an individual patient's background situation" (705).

I am suggesting that we reject the First-Order Consent thesis and instead endorse the Higher-Order Consent thesis, meaning that we can regard higher-order medical information as sufficient for consent being informed. The view goes further than defending the aforementioned right to not know as it is traditionally understood and claims that patients have the right to not know even in contexts where knowing is not possible.[19]

For the rest of this section, I will argue for the Higher-Order Consent thesis on three different grounds. First, I appeal to the purpose of consent. Second, I appeal to theories of disclosure in bioethics. Third, I appeal to the fact that many features of medicine seem to already operate as if the Higher-Order Consent thesis were true.

Before continuing, I should add that the following discussion envisions a world in which patients have the option to be treated by the method of their choice, whether that be opaque AI, explainable AI, or humans. I am not suggesting that endorsing my view requires that we allow only opaque AI to treat patients; I am merely arguing that opaque AI is compatible with informed consent.

i. the purpose of consent:

I'll start with my argument based on the purpose of consent. Broadly speaking, there are two views about what the purpose of consent is, and both support the Higher-Order Consent thesis. According to one view, the purpose of consent is to *protect autonomy*, and according to the other view, the purpose of consent is to *promote wellbeing*.[20] Obviously, consent can do both things at once, but the proponents of these views intend to isolate what is intrinsically important about consent, as the goals of wellbeing and autonomy can easily come apart. I'll argue that both views of the purpose of consent support the view that patients may opt for treatment from an in-principle opaque AI.

Suppose that the purpose of consent is to protect our autonomy by allowing competent and free adults make the choices that they desire to make. As stipulated, we are assuming that treatment via in-principle opaque AI is in the medical best interests of the patient. If patients' goal is to pick the treatment with the highest possible chance for the best medical outcome (which is certainly not

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[19] I should be clear that opponents of the traditional right to not know will not be persuaded by my view. I'm fine with that implication, as I'm not swayed by their concerns. I think objections to the right to not know misunderstand the point of consent. Opponents of the right to not know argue that patients are obligated to know all of the information so that they can choose autonomously; on my view, the point of consent is to allow patients to make the choices they want to make. At the point where we are saying that consent only permits certain kinds of decisions by competent adults, we are missing the point of consent's autonomy protecting role. Arguments against the right to not know in contexts where using this right would harm third parties are compatible with my view. Ost claims that "the right to be informed...is a mandatory right-i.e., it is not an option right which one may or may not exercise; rather it is a right which we are obligated to exercise" (307). As Harris and Keywood argue, "where I give someone (against their will) reliable information about themselves or their condition which is relevant to decisions they must make I may violate a liberty they assert but I do not violate their autonomy, for the information I give them is necessary for their autonomous decision making" (419).

[20] As Beauchamp and Childress say, "since the mid-1970s, the primary justification advanced for requirements of informed consent has been to protect autonomous choice" (77). Elsewhere, Beauchamp puts it like this: "the purpose of consent provisions is not protection from risk, as many earlier federal policies seemed to imply, but the protection of autonomy and personal dignity" (9). Other philosophers have taken the view that the point of consent is to promote wellbeing. As Stacey Taylor argues, "the ethical foundation of informed consent is really concern for human well-being" (384).

everyone's goal), then such patients will want to be treated by opaque AI and would be fine with receiving only higher-order medical information. If we were to deny such individuals this option, we frustrate the point of consent. In the name of informed consent, we should not tell patients that they lack the autonomy to waive certain rights when they deem it beneficial to do so. If the purpose of consent is to promote autonomy, it is disrespectful to autonomy to prevent a competent patient from pursuing an option they believe to be better for them. Again, I'm not envisioning a world in which anyone is being forced to be treated by an AI; human doctors are still available.[21]

If the purpose of consent is to promote wellbeing, and if it's correct that treatment via opaque AI produces the best medical outcome for patients, then the argument goes through quite easily. Allowing treatment via opaque AI would enable patients to achieve the function of consent by doing what is likely to best advance their wellbeing. Again, not allowing treatment via opaque AI would actively frustrate the goal of consent, as it this would hinder an agent from maximizing their own wellbeing.

Thus, on both possible views of the purpose of consent, treatment via opaque medical AI can satisfy the function of consent. And, preventing such a treatment option would actually frustrate the purpose of the consent!

ii. theories of disclosure:

Now, I argue that according to the plausible theories of what disclosure for informed consent requires, treatment via in-principle opaque AI is compatible with informed consent.

According to Beauchamp and Childress, there are three main views about disclosure: (1) the professional practice standard, (2) the reasonable person standard, and (3) the subjective standard. According to the professional practice standard, the relevant information that doctors must disclose is determined by the "professional community's customary practices," meaning that "professional custom establishes the amount and kinds of information to be disclosed" (Beauchamp and Childress 82). Thus, whatever it is common practice to disclose is what doctors are obligated to disclose. According to the reasonable person standard, "we must determine the information to be disclosed by reference to a hypothetical reasonable person" (Beauchamp and Childress). If this reasonable person were to judge a piece of information to be relevant, then that information is relevant. Conversely, if this reasonable person judges a piece of information to not be relevant to their decision, then doctors are not obligated to disclose that information. Lastly, the subjective standard "judges adequacy of information by reference to the specific informational needs of the individual person, rather than the hypothetical 'reasonable person'" (Beauchamp and Childress 83). In short, whether a piece of information is relevant to a patient is determined by the patient herself, but doctors are only obligated to disclose information that either the patient asks for or which the doctor could reasonably think is relevant for this specific patient.

I take it that the professional practice standard is implausible, as it seems to commit a clear is-ought error of inferring from the way clinicians act to the way clinicians ought to act. The two plausible views are the reasonable person standard and the subjective standard. Both views have the implication that informed consent is compatible with being treated and diagnosed by an opaque AI.

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[21] My view is in concert with Ploug and Holm's argument that patients have the right to not be treated or diagnosed by AI.

A reasonable patient would want to receive the best care, regardless of how much first-order information they receive. If they are aware that AI care is most likely to achieve that, then a reasonable person would likely waive their right to an explanation so that they can receive the best medical care available. Suppose a patient has two options:

> **Treatment 1:** she can have a first-order explanation of the treatment and diagnosis, but she has compelling reason to think that there are better treatments available to her.
>
> **Treatment 2:** she cannot have a first-order explanation of the treatment and diagnosis, but there is compelling reason to think that this treatment will be more effective than treatment 1.

Clearly, Treatment 2 is the more rational option to pick and thus would be selected by the reasonable patient. Consider the previously mentioned use of AI to diagnose diabetic retinopathy. AI is substantially better than an ophthalmologist at identifying diabetic retinopathy. The AI (on 819 patients) was 87.2 sensitive (meaning able to correctly identify cases where the person does not have the condition) and 90.7% specific (meaning able to identify cases where the person does have the condition). This dramatically outperforms humans: "Previous studies have shown that board-certified ophthalmologists that perform indirect ophthalmoscopy achieve an average sensitivity of 33%, 34%, or 73%" (Abràmoff et al., 3). If the accuracy of the AI is truly that much better than that of human doctors, it seems plausible that a reasonable person would prefer the AI.

The fact that Treatment 2 seems rational suggests to me that most objections to AI in the clinical setting are really about a fear that the AI is not reliable, not about a concern that it jeopardizes informed consent. In other words, the consent-related concerns that many people have about using AI in medicine seemingly can't be explained by a real concern for consent; instead, the concerns we have about clinical AI are better explained by a background intuition that medical AI isn't actually better than doctors.

As a subjective matter, we'd have to ask individual patients if, given the facts about medical AI, they would prefer treatment via AI (without a first-order explanation) or from a human (with a first-order explanation). According to the subjective standard, what information a patient needs for consent is whatever information they want. If a patient wants only higher-order information, then their consent can be informed with only that information.

In short, both plausible views about what information ought to be disclosed to patients support the Higher-Order Consent thesis.

iii. the status quo supports the higher-order consent thesis:

Although my view conflicts with our rhetoric around informed consent, I'm not sure it's actually in conflict with already well-accepted medical practices. The mechanics of diagnostic tests are not usually explained to patients, and patients don't know how these tests reach a conclusion. They are told the outcome of the test. Those tests are not black boxes, but they may as well be to the patient. What's relevant to the patient is that they are accurate.[22] We don't usually have our doctor's success rate with us as we choose. So, with AI, we would have a piece of information we don't have with the

[22] I draw inspiration for this point from (Amman et al. 4-5).

doctor.[23] Patients often don't ask, nor would they always understand, the doctor's reasoning toward their diagnosis/treatment. But, so long as they are informed about the doctor's reliability, then it seems like they are sufficiently informed to give valid consent.

An objection by analogy enters here. One might object that using opaque AI in clinical medicine would justify absurd practices, such as a doctor using a Magic 8-Ball to make diagnosis and treatment decisions. Like the AI, the 8-Ball is entirely opaque to us. Clearly, current medical standards would not allow this. However, this analogy is easily set aside; as Cohen points out:

> With the Magic 8-Ball or Astrology...not only can the physician not explain why it works, but she also has no epistemic warrant that it works. Conversely, with medical AI/ ML (especially its more opaque forms), the argument is that it is more like Aspirin, in that the former may be true but the latter is not—assuming that physicians have good reason to believe the AI/ML is likely to lead to better decisions. The epistemic warrant for that proposition need not be firsthand knowledge—we might think of medical AI/ML as more like a credence good, where the epistemic warrant is trust in someone else...this is true of most FDA-approved pharmaceuticals. The physician is likely quite ignorant of the underlying trial design or results that led FDA to believe that the drug was safe and effective, but her knowledge that it has been FDA-approved supplies the necessary epistemic warrant (1444).

In short, the analogy fails, because we have no reason to believe that the recommendations of 8-Balls are likely to produce good medical outcomes. If, instead, it were an opaque 8-Ball that had shown itself to be highly reliable in this domain, it seems much more reasonable to use it. Cohen suggests that this is more or less our situation with regard to many common pharmaceutical interventions; we know that they work but don't know why. London makes this point quite powerfully in his defense of opaque AI in medicine. In brief, London argues that just as AI can be opaque but effective, so too are many medical interventions:

> Although medicine is one of the oldest productive sciences, its knowledge of underlying causal systems is in its infancy; the pathophysiology of disease is often uncertain, and the mechanisms through which interventions work is either not known or not well understood. As a result, decisions that are atheoretic, associationist, and opaque are commonplace in medicine...As counterintuitive and unappealing as it may be, the opacity, independence from an explicit domain model, and lack of causal insight associated with some of the most powerful machine learning approaches are not radically different from routine aspects of medical decision-making. Our causal knowledge is often fragmentary, and uncertainty is the rule rather than the exception. In such cases, careful empirical validation of an intervention's practical merits is the most important task (17-18).

Patients already give valid, informed consent to doctors based high-order information alone. As long as the patient knows that the doctor is credentialed, reliable in the past, etc., this seems perfectly fine. Allowing the use of opaque medical AI would merely be an extension of this existing practice. Patients can know the reliability of the AI, how it works, etc.

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[23] As Kwa says, "information about the past performance or reliability of the human medical expert carrying out a treatment (or, less significantly, diagnosis) is not legally required for informed consent," but "such information is in fact available for epistemically opaque AI systems through empirical validation" (910-911).

To sum up my argument, the purpose of consent, theories of disclosure, and the status quo all support the Higher-Order Consent thesis. If this thesis is true, then treatment and diagnosis from in-principle opaque medical AI is compatible with informed consent. My suggestion is that willing patients who are aware of the reliability of medical AI are making consensual decisions when they agree to waive their right to a first-order explanation of their diagnosis and treatment. To suggest that consent requires explainability of first-order information about diagnosis and treatment is to insist that patients must settle for worse medical outcomes.[24]

III. Objections:

i. modal explanations?

One might object that an important difference between the status quo and the potential AI future is that, currently, patients can ask for an explanation. Whereas, in the AI world, this is not even an option. In a sense, the status quo allows for modal explanations, which are explanations a patient could ask for even if they don't do so. Modal explanations are not possible under the opaque AI system. As we discussed earlier in the paper, informed consent does not require that a patient understands the diagnosis, but it may require that the patient *can* have an explanation if they want one. In other words, the patient must have the ability to receive an explanation, even if they ultimately waive this ability. On this standard, opaque AI does not pass, while human doctors do.

I have two responses. First, it's far from true that doctors are able to give a fully explainable line of reasoning for how they reach their decisions, even if patients ask. There's a sense in which the demand for explainability in medicine is not motivated for doctors or AI. We don't typically require that doctors be fully explainable to patients, nor do we expect patients to be able to even understand the information if it were explained to them. Additionally, all human minds are opaque to a certain degree.[25] Second, if the choice is between better outcomes and a patient's right to a modal explanation, it seems clear that patients will want better outcomes. Respect for informed consent requires that we respect their decision. If patients choose to receive treatment that makes modal explanations not possible, this is their autonomous choice. At the risk of repetition, we should not allow our commitment to informed consent to prevent competent adults from seeking high quality care that they want.

ii. pigeons and trust?

I'll close this section with a discussion of a fascinating objection proposed by Alvarado. Alvarado cites evidence which suggests that pigeons can be trained to be highly reliable at radiology; he then uses this as an analogy for why we ought to prefer explainable AI over opaque AI, even if the latter is more accurate:

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[24] London echoes this point: "Any preference for less accurate models...carries risks to patient health and welfare. Without concrete assurance that these risks are offset by the expectation of additional benefits to patients, a blanket preference for simpler models is simply a lethal prejudice" (18).

[25] Some (e.g. Kawamleh, 903) have gone as far as to say that human minds are black boxes. That seems too strong. Even if humans lack full transparency, we can give a rough explanation of our thought process in a way that is not possible with opaque AI.

> Few would feel as confident calling for a significant role for pigeons in assisting in diagnosis, let alone calling for the replacement of radiologists with pigeons. This is not just because of some irrational prejudice towards pigeons, but rather because there is something highly suspect, I argue, about replacing human expertise with an opaque, albeit accurate, methodology simply on the basis of predictive success (123-124).

Alvarado argues that this is the case because "in the context of epistemically demanding inquiry (in which life-critical disciplines...), non-epistemic reasons [practical reasons] are simply not good reasons for the adoption of novel artifacts (132). Instead, Alvarado suggests that "when it comes to the introduction of novel technologies and the trust one can allocate to them within the context of formal inquiry, such as scientific endeavors (in which one seeks to understand and not merely to observe)" (132), we should appeal to epistemic reasons (reasons to believe a proposition is true). As he says, "citing the many lives that could be saved by accepting a proposition, or the efficiency of doing so, does not address the proposition's relation to truth and therefore does not constitute an epistemic reason to adopt it" (132).

I confess that I find this puzzling. I agree that, ideally, we should be able to know why a new medical device works. However, if the tradeoff is between having this knowledge or saving lives, I would choose the latter. I'm willing to concede that this means patients might not have strong reasons to trust opaque AI, but this perhaps shows that patients have the autonomy to receive treatment from sources that they, strictly speaking, don't trust.[26] Perhaps the catchier way to put it is that if pigeon radiologists reliably produced more accurate results than human radiologists, I say bring on the pigeon radiologists! Additionally, I'll note that while the goal of medical research is to find the truth, the goal of patients is purely pragmatic: to achieve the best medical outcome. Thus, if we agree that medicine is not only a truth-seeking enterprise, then I'm not sure it's a problem for pragmatic reasons to justify the adoption of a novel medical technology.

V. Conclusion

I've argued that medical AI (even if it's in-principle opaque) does not threaten informed consent in clinical medicine. Importantly, the view I defend in this paper is compatible with continuing to prioritize research into explainable AI. All I mean to defend is that informed consent and opaque AI are compatible, but this is consistent with informed consent being better achieved by explainable AI. It would clearly be the best option if we could have highly accurate and explainable AI in medicine.

We might reasonably wonder what medicine would look like were this view to be adopted. The goal of this paper is not to work out the logistical details of how this might all play out in a clinical setting. However, I would be remiss if I didn't address this issue. Some may worry that what I'm suggesting is a dangerous return to the old paternalism that characterized medicine prior to the informed consent movement. Fortunately, this is not the case. There's nothing paternalistic about my proposal, as patient autonomy is the deciding factor all the way down. It is patients who are waiving a right to a first-order explanation, not doctors mandating this or withholding it from them when they would otherwise want to receive it. Several outcomes are possible, from completely replacing doctors with AI to using AI merely as a diagnostic aid. My own view is that the outcome with no human doctors remaining in medicine is not a good one. Although, such an outcome is compatible

[26] For those who find this too concessive, Durán and Jongsma have argued that "the reliability of algorithms provides reasons for trusting the outcomes of medical artificial intelligence" (329).

with respect for informed consent, so long as everyone genuinely prefers this outcome. But, consent is not the only value in medicine. We want patients to be comfortable, which is likely better achieved with humans still involved in the process. Perhaps my view would open the door to moving from, "patients may pursue AI care consistent with informed consent," to "patients can give informed consent to an AI, so let's only have AI doctors from here on." I do not think this is likely to happen. There will certainly still be demand for human doctors. Many people will not trust AI, and many will think that it's not really more reliable than human doctors.[27] In a market system, that demand will furnish a supplier. However, if the demand for human doctors decreases sufficiently, we may reach a point where a market no longer furnishes them. On my view, such an outcome is compatible with informed consent, but individuals who still insist on exercising their right to first-order explanation of their treatment would be entitled to treatment via human doctors. As I've gone to great pains to emphasize, what patients receive in terms of information should be respectful of what they want to receive. Just as insisting that they must receive only first-order information does not respect their consent, so too does insisting that they can only receive higher-order information. In that regard, I differ from Cohen, who claims that "if using AI/ML really produced better patient outcomes across the board, then it seems desirable for it to become the standard of care. As a matter of informed consent, ethically or legally, it is not clear why we should shed tears if in such a world, patients do not have access to the non-AI/ML approach" (1463).

[27] Evidence suggests that patients will be hesitant to use medical AI: "large-scale adoption of AI hinges not only on adoption by healthcare systems and providers but also on patient utilization, and patients are reluctant to utilize medical AI" (Cadario, Longoni, and Morewedge 1636).

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